



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | rudder-controller  |
| Board Name      | NUCLEO-U575ZI-Q    |
| Generated with: | STM32CubeMX 6.14.0 |
| Date            | 11/01/2025         |

### 1.2. MCU

|                |                |
|----------------|----------------|
| MCU Series     | STM32U5        |
| MCU Line       | STM32U575/585  |
| MCU name       | STM32U575ZITxQ |
| MCU Package    | LQFP144        |
| MCU Pin number | 144            |

### 1.3. Core(s) information

|         |                |
|---------|----------------|
| Core(s) | ARM Cortex-M33 |
|---------|----------------|



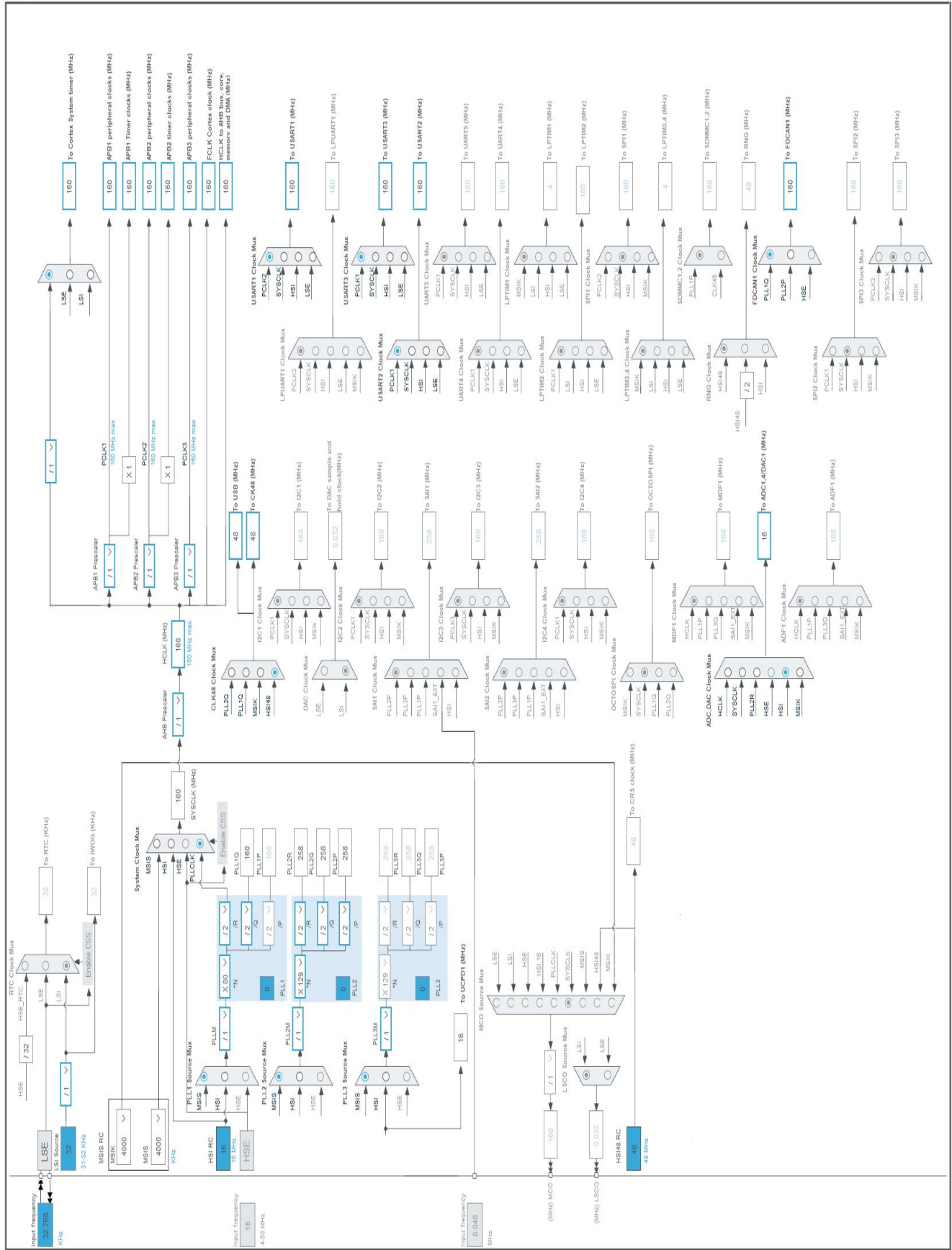
### 3. Pins Configuration

| Pin Number<br>LQFP144 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label       |
|-----------------------|---------------------------------------|----------|--------------------------|-------------|
| 6                     | VBAT                                  | Power    |                          |             |
| 7                     | PC13                                  | I/O      | GPIO_EXTI13              | USER_BUTTON |
| 8                     | PC14-OSC32_IN (PC14)                  | I/O      | RCC_OSC32_IN             |             |
| 9                     | PC15-OSC32_OUT (PC15)                 | I/O      | RCC_OSC32_OUT            |             |
| 16                    | VSS                                   | Power    |                          |             |
| 17                    | VDD                                   | Power    |                          |             |
| 19                    | PF7                                   | I/O      | FDCAN1_RX                |             |
| 20                    | PF8                                   | I/O      | FDCAN1_TX                |             |
| 25                    | NRST                                  | Reset    |                          |             |
| 28                    | PC2                                   | I/O      | ADC1_IN3                 | VBUS_SENSE  |
| 30                    | VSSA                                  | Power    |                          |             |
| 32                    | VDDA                                  | Power    |                          |             |
| 35                    | PA2                                   | I/O      | USART2_TX                |             |
| 36                    | PA3                                   | I/O      | USART2_RX                |             |
| 37                    | VSS                                   | Power    |                          |             |
| 38                    | VDD                                   | Power    |                          |             |
| 48                    | VSS                                   | Power    |                          |             |
| 49                    | VDD                                   | Power    |                          |             |
| 58                    | VSS                                   | Power    |                          |             |
| 59                    | VDD                                   | Power    |                          |             |
| 68                    | VLXSMPS                               | Power    |                          |             |
| 69                    | VDDSMPS                               | Power    |                          |             |
| 70                    | VSSSMPS                               | Power    |                          |             |
| 71                    | VDD11                                 | Power    |                          |             |
| 72                    | VSS                                   | Power    |                          |             |
| 73                    | VDD                                   | Power    |                          |             |
| 75                    | PB14 *                                | I/O      | GPIO_Input               | UCPD_FLT    |
| 76                    | PB15                                  | I/O      | UCPD1_CC2                | UCPD1_CC2   |
| 83                    | VSS                                   | Power    |                          |             |
| 84                    | VDD                                   | Power    |                          |             |
| 87                    | PG2 *                                 | I/O      | GPIO_Output              | LED_RED     |
| 94                    | VSS                                   | Power    |                          |             |
| 95                    | VDDIO2                                | Power    |                          |             |
| 97                    | PC7 *                                 | I/O      | GPIO_Output              | LED_GREEN   |
| 101                   | PA9                                   | I/O      | USART1_TX                | USART1_TX   |
| 102                   | PA10                                  | I/O      | USART1_RX                | USART1_RX   |

| Pin Number<br>LQFP144 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label         |
|-----------------------|---------------------------------------|----------|--------------------------|---------------|
| 103                   | PA11                                  | I/O      | USB_OTG_FS_DM            | USB_OTG_FS_DM |
| 104                   | PA12                                  | I/O      | USB_OTG_FS_DP            | USB_OTG_FS_DP |
| 105                   | PA13 (JTMS/SWDIO)                     | I/O      | DEBUG_JTMS-SWDIO         |               |
| 106                   | VDDUSB                                | Power    |                          |               |
| 107                   | VSS                                   | Power    |                          |               |
| 108                   | VDD                                   | Power    |                          |               |
| 109                   | PA14 (JTCK/SWCLK)                     | I/O      | DEBUG_JTCK-SWCLK         |               |
| 110                   | PA15 (JTDI)                           | I/O      | UCPD1_CC1                | UCPD1_CC1     |
| 111                   | PC10                                  | I/O      | USART3_TX                |               |
| 112                   | PC11                                  | I/O      | USART3_RX                |               |
| 120                   | VSS                                   | Power    |                          |               |
| 121                   | VDD                                   | Power    |                          |               |
| 129                   | VSS                                   | Power    |                          |               |
| 130                   | VDDIO2                                | Power    |                          |               |
| 132                   | PB3 (JTDO/TRACESWO)                   | I/O      | DEBUG_JTDO-SWO           |               |
| 134                   | PB5 *                                 | I/O      | GPIO_Output              | UCPD_DBn      |
| 136                   | PB7 *                                 | I/O      | GPIO_Output              | LED_BLUE      |
| 142                   | VDD11                                 | Power    |                          |               |
| 143                   | VSS                                   | Power    |                          |               |
| 144                   | VDD                                   | Power    |                          |               |

\* The pin is affected with an I/O function

## 4. Clock Tree Configuration



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |                |
|-----------|----------------|
| Series    | STM32U5        |
| Line      | STM32U575/585  |
| MCU       | STM32U575ZITxQ |
| Datasheet | DS13737 Rev1   |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

#### 1.4. Sequence

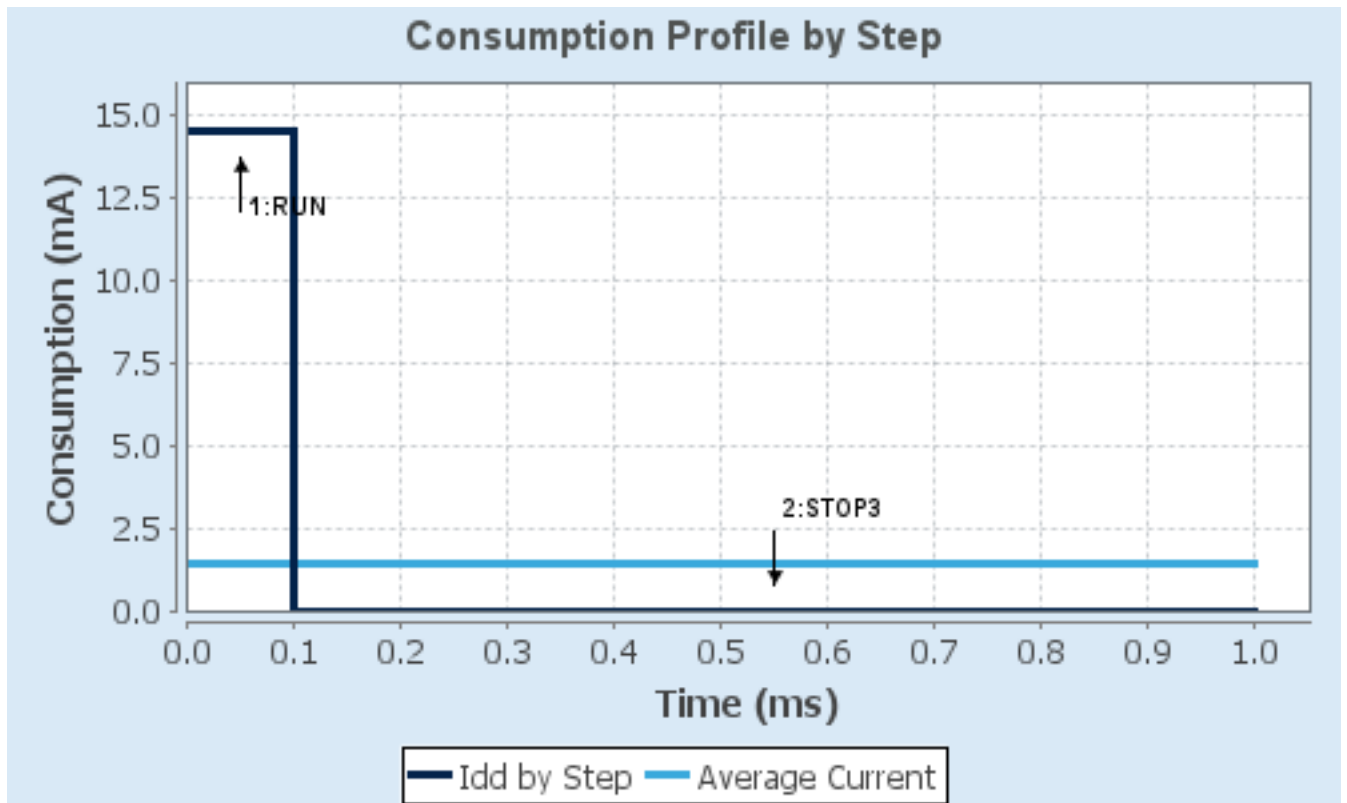
|                               |                                      |                |
|-------------------------------|--------------------------------------|----------------|
| <b>Step</b>                   | Step1                                | Step2          |
| <b>Mode</b>                   | RUN                                  | STOP3          |
| <b>Vdd</b>                    | 3.0                                  | 3.0            |
| <b>Voltage Source</b>         | Battery                              | Battery        |
| <b>Range</b>                  | Range1-High                          | NoScale/SMPS   |
| <b>Fetch Type</b>             | FLASH_PwrDwnBank2/ART/<br>Cache2Ways | FLASH          |
| <b>CPU Frequency</b>          | 160 MHz                              | 0 Hz           |
| <b>Clock Configuration</b>    | HSE BYP PLL<br>ALL_RAM_RETENTION     | ALL_CLOCKS_OFF |
| <b>Clock Source Frequency</b> | 16 MHz                               | 0 Hz           |
| <b>Peripherals</b>            |                                      |                |
| <b>Additional Cons.</b>       | 0 mA                                 | 0 mA           |
| <b>Average Current</b>        | 14.5 mA                              | 1.8 $\mu$ A    |
| <b>Duration</b>               | 0.1 ms                               | 0.9 ms         |
| <b>DMIPS</b>                  | 200.0                                | 0.0            |
| <b>Ta Max</b>                 | 103.43                               | 105            |
| <b>Category</b>               | In DS Table                          | In DS Table    |

#### 1.5. Results

|               |                              |                 |            |
|---------------|------------------------------|-----------------|------------|
| Sequence Time | 1 ms                         | Average Current | 1.45 mA    |
| Battery Life  | 3 months, 6 days,<br>2 hours | Average DMIPS   | 20.0 DMIPS |

#### 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                                                            |
|-----------------------------------|------------------------------------------------------------------|
| Project Name                      | rudder-controller                                                |
| Project Folder                    | C:\Users\lemmak\Sailbot\com-module-firmware\projects\controller- |
| Toolchain / IDE                   | STM32CubeIDE                                                     |
| Firmware Package Name and Version | STM32Cube FW_U5 V1.7.0                                           |
| Application Structure             | Advanced                                                         |
| Generate Under Root               | Yes                                                              |
| Do not generate the main()        | No                                                               |
| Minimum Heap Size                 | 0x200                                                            |
| Minimum Stack Size                | 0x400                                                            |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | No                                    |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name          | Peripheral Instance Name |
|------|------------------------|--------------------------|
| 1    | SystemClock_Config     | RCC                      |
| 2    | MX_GPIO_Init           | GPIO                     |
| 3    | MX_GPDMA1_Init         | GPDMA1                   |
| 4    | MX_ADC1_Init           | ADC1                     |
| 5    | MX_ICACHE_Init         | ICACHE                   |
| 6    | MX_UCPD1_Init          | UCPD1                    |
| 7    | MX_USART1_UART_Init    | USART1                   |
| 8    | MX_FDCAN1_Init         | FDCAN1                   |
| 9    | MX_USART2_UART_Init    | USART2                   |
| 10   | MX_USART3_UART_Init    | USART3                   |
| 11   | MX_USB_OTG_FS_PCD_Init | USB_OTG_FS               |



## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

#### IN3: IN3 Single-ended

##### 3.1.1. Parameter Settings:

###### ADCs\_Common\_Settings:

Mode Independent mode

###### ADC\_Settings:

Clock Prescaler Asynchronous clock mode divided by 1

Resolution ADC 14-bit resolution

Gain Compensation 0

Scan Conversion Mode Disabled

Continuous Conversion Mode Disabled

Discontinuous Conversion Mode Disabled

Trigger Frequency High frequency

End Of Conversion Selection End of single conversion

Overrun behaviour Overrun data preserved

Left Bit Shift No bit shift

Conversion Data Management Mode Regular Conversion data stored in DR register only

Low Power Auto Wait Disabled

###### ADC\_Regular\_ConversionMode:

Enable Regular Conversions Disable

Oversampling Ratio 1

###### ADC\_Injected\_ConversionMode:

Enable Injected Conversions Disable

### 3.2. DEBUG

#### Debug: Trace Asynchronous Sw

### 3.3. FDCAN1

#### mode: Activated

##### 3.3.1. Parameter Settings:

###### Basic Parameters:

Clock Divider Divide kernel clock by 1

Frame Format **FD mode without BitRate Switching \***

Mode **External LoopBack mode \***

|                         |                 |
|-------------------------|-----------------|
| Auto Retransmission     | <b>Enable *</b> |
| Transmit Pause          | Disable         |
| Protocol Exception      | Disable         |
| Nominal Sync Jump Width | <b>3 *</b>      |
| Data Prescaler          | <b>2 *</b>      |
| Data Sync Jump Width    | <b>11 *</b>     |
| Data Time Seg1          | <b>13 *</b>     |
| Data Time Seg2          | <b>11 *</b>     |
| Std Filters Nbr         | <b>1 *</b>      |
| Ext Filters Nbr         | <b>1 *</b>      |
| Tx Fifo Queue Mode      | FIFO mode       |

#### **Bit Timings Parameters:**

|                          |                  |
|--------------------------|------------------|
| Nominal Prescaler        | <b>5 *</b>       |
| Nominal Time Quantum     | <b>31.25 *</b>   |
| Nominal Time Seg1        | <b>16 *</b>      |
| Nominal Time Seg2        | <b>3 *</b>       |
| Nominal Time for one Bit | <b>625 *</b>     |
| Nominal Baud Rate        | <b>1600000 *</b> |

### **3.4. ICACHE**

**mode: Memory address remap**

**Mode: 1-way (direct mapped cache)**

#### 3.4.1. Parameter Settings:

##### **Region 0:**

|        |         |
|--------|---------|
| Region | Disable |
|--------|---------|

##### **Region 1:**

|        |         |
|--------|---------|
| Region | Disable |
|--------|---------|

##### **Region 2:**

|        |         |
|--------|---------|
| Region | Disable |
|--------|---------|

##### **Region 3:**

|        |         |
|--------|---------|
| Region | Disable |
|--------|---------|

### **3.5. LPBAM**

**mode: LPBAM Scenario uses resources from Smart Run Domain only**

**mode: LPBAM Scenario is hosted by LPDMA1**

### 3.6. LPBAMQUEUE

**mode: QUEUE MODE**

#### 3.6.1. Parameter Settings:

##### **DMA Channel Configuration:**

|          |     |
|----------|-----|
| Priority | Low |
|----------|-----|

##### **DMA Channel Interrupt Configuration:**

|                               |         |
|-------------------------------|---------|
| Data Transfer Error Interrupt | Disable |
| Update Link Error Interrupt   | Disable |
| User Setting Error Interrupt  | Disable |
| Transfer Complete Interrupt   | Disable |
| Trigger Overrun Interrupt     | Disable |

### 3.7. MEMORYMAP

**mode: Activated**

### 3.8. PWR

**mode: Dead Battery Signals disabled**

**mode: Power saving mode**

**mode: Privilege attributes**

**mode: Control Vdda isolation**

#### 3.8.1. Power Saving:

##### **System power supply:**

|                 |        |
|-----------------|--------|
| Power Regulator | SMPS * |
|-----------------|--------|

##### **SRAM power down in Run mode:**

|                              |         |
|------------------------------|---------|
| SRAM1 power down in Run mode | Disable |
| SRAM2 power down in Run mode | Disable |
| SRAM3 power down in Run mode | Disable |
| SRAM4 power down in Run mode | Disable |

##### **SRAM power down in Stop mode:**

|                                                  |         |
|--------------------------------------------------|---------|
| SRAM1 Page1 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM1 Page2 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM1 Page3 power down in Stop (0, 1, 2, 3) mode | Disable |

|                                                  |         |
|--------------------------------------------------|---------|
| SRAM2 Page1 power down in Stop (0, 1, 2) mode    | Disable |
| SRAM2 Page2 power down in Stop (0, 1, 2) mode    | Disable |
| SRAM3 Page1 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page2 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page3 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page4 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page5 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page6 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page7 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM3 Page8 power down in Stop (0, 1, 2, 3) mode | Disable |
| SRAM4 power down in Stop (0, 1, 2, 3) mode       | Disable |
| ICACHE power down in Stop (0, 1, 2, 3) mode      | Disable |
| DCACHE1 power down in Stop (0, 1, 2, 3) mode     | Disable |
| DMA2D RAM power down in Stop (0, 1, 2, 3) mode   | Disable |
| PKA32 RAM power down in Stop (0, 1, 2, 3) mode   | Disable |
| PERIPH RAM power down in Stop (0, 1, 2, 3) mode  | Disable |

#### **SRAM fast wakeup:**

|                                                |         |
|------------------------------------------------|---------|
| SRAM4 fast wakeup from Stop (0, 1, 2, 3) modes | Disable |
|------------------------------------------------|---------|

### 3.8.2. PWR Privilege :

#### **Privilege PWR:**

|                                   |         |
|-----------------------------------|---------|
| Privilege of PWR Secure Items     | Disable |
| Privilege of PWR Non-Secure Items | Disable |
| PWR Privilege                     | Disable |

### 3.8.3. PWR Security:

#### **Secure PWR:**

|                                                    |         |
|----------------------------------------------------|---------|
| Wake-Up 1 secure protection                        | Disable |
| Wake-Up 2 secure protection                        | Disable |
| Wake-Up 3 secure protection                        | Disable |
| Wake-Up 4 secure protection                        | Disable |
| Wake-Up 5 secure protection                        | Disable |
| Wake-Up 6 secure protection                        | Disable |
| Wake-Up 7 secure protection                        | Disable |
| Wake-Up 8 secure protection                        | Disable |
| Voltage detection and monitoring secure protection | Disable |
| Pull-up/pull-down secure protection                | Disable |
| Low power modes secure protection                  | Disable |
| Backup domain secure protection                    | Disable |

### 3.9. RCC

#### Low Speed Clock (LSE) : Crystal/Ceramic Resonator

##### 3.9.1. RCC Privilege :

###### Privilege RCC:

|                                   |         |
|-----------------------------------|---------|
| Privilege of RCC Non-Secure Items | Disable |
|-----------------------------------|---------|

##### 3.9.2. Parameter Settings:

###### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Flash Latency(WS) | 4 WS (5 CPU cycle) |

###### RCC Parameters:

|                                |          |
|--------------------------------|----------|
| HSI Calibration Value          | 16       |
| MSI Calibration Value          | 16       |
| MSIS/MSIK Auto Calibration     | Disabled |
| HSE Startup Timeout Value (ms) | 100      |
| LSE Startup Timeout Value (ms) | 5000     |

###### Power Parameters:

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

###### PLL1/2/3 Parameters:

|                                |                     |
|--------------------------------|---------------------|
| PLL1M BOOST EPOD Clock Divider | 1                   |
| PLL1 input frequency range     | Between 4 and 8 MHz |

###### Low Power Parameters:

|                  |          |
|------------------|----------|
| MSI in Stop mode | Disabled |
| HSI in Stop mode | Disabled |

### 3.10. SYS

#### Timebase Source: SysTick

### 3.11. UCPD1

#### UCPD Mode: Sink

##### 3.11.1. Parameter Settings:

|         |     |
|---------|-----|
| Version | 1.0 |
|---------|-----|



### 3.12. USART1

#### Mode: Asynchronous

##### 3.12.1. Parameter Settings:

###### Basic Parameters:

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

###### Advanced Parameters:

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |
| Autonomous Mode  | Disable                     |

###### Advanced Features:

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.13. USART2

#### Mode: Asynchronous

##### 3.13.1. Parameter Settings:

###### Basic Parameters:

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

**Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |
| Autonomous Mode  | Disable                     |

**Advanced Features:**

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.14. USART3

**Mode: Asynchronous**3.14.1. Parameter Settings:**Basic Parameters:**

|             |                           |
|-------------|---------------------------|
| Baud Rate   | 115200                    |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

**Advanced Parameters:**

|                  |                             |
|------------------|-----------------------------|
| Data Direction   | Receive and Transmit        |
| Over Sampling    | 16 Samples                  |
| Single Sample    | Disable                     |
| ClockPrescaler   | 1                           |
| Fifo Mode        | Disable                     |
| Txfifo Threshold | 1 eighth full configuration |
| Rxfifo Threshold | 1 eighth full configuration |
| Autonomous Mode  | Disable                     |

**Advanced Features:**

|                               |         |
|-------------------------------|---------|
| Auto Baudrate                 | Disable |
| TX Pin Active Level Inversion | Disable |

|                               |         |
|-------------------------------|---------|
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX Pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

### 3.15. USB\_OTG\_FS

#### Mode: Device\_Only

##### 3.15.1. Parameter Settings:

|                                     |                     |
|-------------------------------------|---------------------|
| Speed                               | Full Speed 12MBit/s |
| Low power                           | Disabled            |
| Battery charging                    | Disabled            |
| Link Power Management               | Disabled            |
| VBUS sensing                        | Disabled            |
| Use dedicated end point 1 interrupt | Disabled            |
| Signal start of frame               | Disabled            |

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP         | Pin                      | Signal           | GPIO mode                                                          | GPIO pull/up pull down      | Max Speed          | User Label    |
|------------|--------------------------|------------------|--------------------------------------------------------------------|-----------------------------|--------------------|---------------|
| ADC1       | PC2                      | ADC1_IN3         | Analog mode                                                        | No pull-up and no pull-down | n/a                | VBUS_SENSE    |
| DEBUG      | PA13<br>(JTMS/SWDIO)     | DEBUG_JTMS-SWDIO | n/a                                                                | n/a                         | n/a                |               |
|            | PA14<br>(JTCK/SWCLK)     | DEBUG_JTCK-SWCLK | n/a                                                                | n/a                         | n/a                |               |
|            | PB3<br>(JTDO/TRACESWO)   | DEBUG_JTDO-SWO   | n/a                                                                | n/a                         | n/a                |               |
| FDCAN1     | PF7                      | FDCAN1_RX        | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low                |               |
|            | PF8                      | FDCAN1_TX        | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low                |               |
| RCC        | PC14-OSC32_IN<br>(PC14)  | RCC_OSC32_IN     | n/a                                                                | n/a                         | n/a                |               |
|            | PC15-OSC32_OUT<br>(PC15) | RCC_OSC32_OUT    | n/a                                                                | n/a                         | n/a                |               |
| UCPD1      | PB15                     | UCPD1_CC2        | Analog mode                                                        | No pull-up and no pull-down | n/a                | UCPD1_CC2     |
|            | PA15 (JTDI)              | UCPD1_CC1        | Analog mode                                                        | No pull-up and no pull-down | n/a                | UCPD1_CC1     |
| USART1     | PA9                      | USART1_TX        | Alternate Function Push Pull                                       | <b>Pull-down *</b>          | <b>Very High *</b> | USART1_TX     |
|            | PA10                     | USART1_RX        | Alternate Function Push Pull                                       | <b>Pull-down *</b>          | <b>Very High *</b> | USART1_RX     |
| USART2     | PA2                      | USART2_TX        | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low                |               |
|            | PA3                      | USART2_RX        | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low                |               |
| USART3     | PC10                     | USART3_TX        | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low                |               |
|            | PC11                     | USART3_RX        | Alternate Function Push Pull                                       | No pull-up and no pull-down | Low                |               |
| USB_OTG_FS | PA11                     | USB_OTG_FS_DM    | Alternate Function Push Pull                                       | No pull-up and no pull-down | <b>Very High *</b> | USB_OTG_FS_DM |
|            | PA12                     | USB_OTG_FS_DP    | Alternate Function Push Pull                                       | No pull-up and no pull-down | <b>Very High *</b> | USB_OTG_FS_DP |
| GPIO       | PC13                     | GPIO_EXTI13      | <b>External Interrupt Mode with Falling edge trigger detection</b> | No pull-up and no pull-down | n/a                | USER_BUTTON   |
|            | PB14                     | GPIO_Input       | Input mode                                                         | No pull-up and no pull-down | n/a                | UCPD_FLT      |
|            | PG2                      | GPIO_Output      | Output Push Pull                                                   | <b>Pull-up *</b>            | <b>High *</b>      | LED_RED       |

| IP | Pin | Signal      | GPIO mode        | GPIO pull/up pull down      | Max Speed     | User Label |
|----|-----|-------------|------------------|-----------------------------|---------------|------------|
|    | PC7 | GPIO_Output | Output Push Pull | <b>Pull-up *</b>            | <b>High *</b> | LED_GREEN  |
|    | PB5 | GPIO_Output | Output Push Pull | No pull-up and no pull-down | Low           | UCPD_DBn   |
|    | PB7 | GPIO_Output | Output Push Pull | <b>Pull-up *</b>            | <b>High *</b> | LED_BLUE   |

## 4.2. GPDMA1

### Channel 15 - 8 Words Internal FIFO / 2D addressing: Standard Request Mode

#### 4.2.1. All Channels:

##### Channel 15:

Request GPDMA1\_REQUEST\_USART3\_RX

#### 4.2.2. SECURITY:

##### CH15:

Enable Channel as Privileged NON PRIVILEGED

#### 4.2.3. CH15:

##### Circular configuration:

Circular Mode Disable

##### Request Configuration:

Request **USART3\_RX \***

DMA Handle in IP Structure hdmarx

Block HW request protocol Single/Burst Level

##### Channel configuration:

Priority Low

Transaction Mode Normal

Direction Peripheral To Memory

##### Source Data Setting:

Source Address Increment After Transfer Disabled

Data Width Byte

Burst Length 1

Allocated Port for Transfer Port 0

##### Destination Data Setting:

Destination Address Increment After Transfer **Enabled \***

Data Width Byte

|                                      |                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------|
| Burst Length                         | 1                                                                                   |
| Allocated Port for Transfer          | Port 0                                                                              |
| <b>Data Handling:</b>                |                                                                                     |
| Data Handling Configuration          | Disable                                                                             |
| <b>Trigger:</b>                      |                                                                                     |
| Trigger Configuration                | Disable                                                                             |
| <b>2D Addressing:</b>                |                                                                                     |
| Configuration                        | Disabled                                                                            |
| <b>Transfer Event Configuration:</b> |                                                                                     |
| Transfer Event Generation            | The TC (and the HT) event is generated at the (respectively half) end of each block |

### 4.3. LINKEDLIST

### 4.4. LPDMA1

## 4.5. NVIC configuration

### 4.5.1. NVIC

| Interrupt Table                         | Enable | Preenmption Priority | SubPriority |
|-----------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                  | true   | 0                    | 0           |
| Hard fault interrupt                    | true   | 0                    | 0           |
| Memory management fault                 | true   | 0                    | 0           |
| Prefetch fault, memory access fault     | true   | 0                    | 0           |
| Undefined instruction or illegal state  | true   | 0                    | 0           |
| System service call via SWI instruction | true   | 0                    | 0           |
| Debug monitor                           | true   | 0                    | 0           |
| Pendable request for system service     | true   | 0                    | 0           |
| System tick timer                       | true   | 0                    | 0           |
| FDCAN1 interrupt 0                      | true   | 0                    | 0           |
| FDCAN1 interrupt 1                      | true   | 0                    | 0           |
| USART3 global interrupt                 | true   | 0                    | 0           |
| USB OTG FS global interrupt             | true   | 0                    | 0           |
| GPDMA1 Channel 15 global interrupt      | true   | 0                    | 0           |
| Flash non-secure global interrupt       | unused |                      |             |
| RCC non-secure global interrupt         | unused |                      |             |
| EXTI Line13 interrupt                   | unused |                      |             |
| ADC1 global interrupt                   | unused |                      |             |
| USART1 global interrupt                 | unused |                      |             |
| USART2 global interrupt                 | unused |                      |             |
| FPU global interrupt                    | unused |                      |             |
| UCPD1 global interrupt                  | unused |                      |             |
| Instruction cache global interrupt      | unused |                      |             |

### 4.5.2. NVIC Code generation

| Enabled interrupt Table                 | Select for init sequence ordering | Generate IRQ handler | Call HAL handler |
|-----------------------------------------|-----------------------------------|----------------------|------------------|
| Non maskable interrupt                  | false                             | true                 | false            |
| Hard fault interrupt                    | false                             | true                 | false            |
| Memory management fault                 | false                             | true                 | false            |
| Prefetch fault, memory access fault     | false                             | true                 | false            |
| Undefined instruction or illegal state  | false                             | true                 | false            |
| System service call via SWI instruction | false                             | true                 | false            |
| Debug monitor                           | false                             | true                 | false            |
| Pendable request for system service     | false                             | true                 | false            |
| System tick timer                       | false                             | true                 | true             |
| FDCAN1 interrupt 0                      | false                             | true                 | true             |

| Enabled interrupt Table            | Select for init<br>sequence ordering | Generate IRQ<br>handler | Call HAL handler |
|------------------------------------|--------------------------------------|-------------------------|------------------|
| FDCAN1 interrupt 1                 | false                                | true                    | true             |
| USART3 global interrupt            | false                                | true                    | true             |
| USB OTG FS global interrupt        | false                                | true                    | true             |
| GPDMA1 Channel 15 global interrupt | false                                | true                    | true             |

**\* User modified value**



## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

| Middleware   |        |        |              |            |          |           |                 |                   |            |     |       |
|--------------|--------|--------|--------------|------------|----------|-----------|-----------------|-------------------|------------|-----|-------|
| System Core  | Analog | Timers | Connectivity | Multimedia | Security | Computing | Trace and Debug | Power and Thermal | Utilities  | Bsp | Other |
| CORTEX_M33 ✓ | ADC1 ✓ |        | FDCAI1 ✓     |            |          |           | DEBUG ✓         | PWR ✓             | LINKEDLIST |     |       |
| GPDMA1 ✓     |        |        | UCPD1 ✓      |            |          |           |                 |                   |            |     |       |
| GPIO ✓       |        |        | USART1 ✓     |            |          |           |                 |                   |            |     |       |
| ICACHE ✓     |        |        | USART2 ✓     |            |          |           |                 |                   |            |     |       |
| LPDMA1       |        |        | USART3 ✓     |            |          |           |                 |                   |            |     |       |
| IVVIC ✓      |        |        | USB_FS ✓     |            |          |           |                 |                   |            |     |       |
| RCC ✓        |        |        |              |            |          |           |                 |                   |            |     |       |
| SYS ✓        |        |        |              |            |          |           |                 |                   |            |     |       |

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                                                      |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32u5_bsd.zip">https://www.st.com/resource/en/bsdl_model/stm32u5_bsd.zip</a>                                                                                                                                                                                         |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32u5-ibis.zip">https://www.st.com/resource/en/ibis_model/stm32u5-ibis.zip</a>                                                                                                                                                                                       |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32u5_svd.zip">https://www.st.com/resource/en/svd/stm32u5_svd.zip</a>                                                                                                                                                                                                       |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                                                       |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                                               |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32u5-series-product-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32u5-series-product-overview.pdf</a>                                                                                           |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32u5-mcu-lines-for-advanced-graphics.pdf">https://www.st.com/resource/en/product_presentation/stm32u5-mcu-lines-for-advanced-graphics.pdf</a>                                                                                                             |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/brstm32ulp.pdf">https://www.st.com/resource/en/brochure/brstm32ulp.pdf</a>                                                                                                                                                                                               |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                                               |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32u5.pdf">https://www.st.com/resource/en/flyer/flstm32u5.pdf</a>                                                                                                                                                                                                       |
| Security Bulletin       | <a href="https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf">https://www.st.com/resource/en/security_bulletin/sb0023-eucleak-protection-statement-for-stmicroelectronics-certified-products-stmicroelectronics.pdf</a> |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf</a>                                                 |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf</a>                                                     |
| Application Notes       | <a href="https://www.st.com/resource/en/application_note/an3126-audio-and-">https://www.st.com/resource/en/application_note/an3126-audio-and-</a>                                                                                                                                                                         |

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& Software

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